Azure SQL Server – Decision Tree for Architects

Choose the Right Model, Tier, and Licensing Strategy

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START

What type of database workload are you running?

FULLY MANAGED PaaS

I want minimal management of infrastructure

→ **Choose Azure SQL (Database / MI)**

Which deployment option fits your needs best?

Azure SQL Database (Single Database)

- Best for most applications
- Built-in high availability, backups, and security
- Up to 100 TB (Hyperscale)
- Fully elastic and scalable

Azure SQL Managed Instance (Instance-level)

- Need instance-level features e.g., SQL Agent, cross-DB queries
- Lift-and-shift (minimal code changes)
- VNet integration, near 100% compatibility with SQL Server
- Up to 16 TB

Which compute model do you prefer for Azure SQL Database or MI?

DTU Model (Simple, all-in-one)

- Predictable monthly cost
- Compute + Storage + IOPS bundled
- Best for databases ≤ 4 TB
- No Azure Hybrid Benefit

vCore Model – Provisioned (Flexible & Transparent)

- Choose exact vCores
- Compute and storage billed separately
- Full visibility of CPU, RAM, IOPS
- Supports Azure Hybrid Benefit
- Best for production workloads

vCore Model – Serverless (Consumption-based)

- Auto-scale between min and max vCores
- Pay per second of actual usage
- Auto-pause when idle
- No Azure Hybrid Benefit
- Best for intermittent or unpredictable workloads

Which service tier is right for your workload (in vCore models)?

General Purpose (GP)

- Balanced compute and storage
- For most OLTP applications
- Up to 80 vCores
- Up to 4 TB storage

Business Critical (BC)

- Highest performance and availability
- In-memory OLTP
- Always On, zone redundant
- 1 free readable replica
- Up to 128 vCores
- Up to 4 TB storage

Hyperscale (HS)

- Massive scale and high IOPS
- For VLDBs and analytics
- Multiple read replicas
- Up to 80+ vCores
- Up to 100 TB storage

Do you have existing SQL Server licenses with Software Assurance?

YES → Use Azure Hybrid Benefit (AHB)
(Available only on vCore – Provisioned)

- Bring your licenses and save up to 30%+
- Combine with Reserved Instances for up to 55%+ savings

NO → Use License Included Pricing
(Pay for SQL license in Azure)

- Simple pricing
- No license management

FULL CONTROL IaaS

I need OS-level access or custom configurations

→ **Choose Azure SQL on Azure Virtual Machines**

Do you need full OS access, custom configurations, or third-party software installation?

NO → Azure SQL Managed Instance

YES → Azure SQL on Azure Virtual Machines

- Full control over OS and SQL Server
- Choose Windows Server and SQL version
- Design your own HA/DR (Always On, FCI, etc.)
- Use BYOL for cost savings
- Manage backups, patching and updates

Choose the Right VM & Storage

Compute (VM Families)

- D / E / Fsv2 – General purpose
- Edsv5 / Edsw5 – Memory optimized
- Lsv3 – Storage optimized
- Msv2 – Balanced workloads

Storage (Disks)

- Premium SSD (P-series) – High performance
- Ultra Disk – Highest performance
- Standard SSD (E-series) – Cost effective
- Use Storage Spaces / RAID as needed

Design High Availability

- Always On Availability Groups
- Failover Cluster Instances (FCI)
- Windows Server Failover Clustering
- Multi-Subnet, Load Balancer
- Backup to Azure Blob / Managed Disks

HYBRID / EDGE SCENARIOS

I need hybrid benefits or specialized deployment

→ Evaluate **Azure Arc enabled SQL / Managed Instance**

Do you need fo run SQL closer to on-premises or across hybrid environments?

YES → Azure SQL Managed Instance (With VNet / Private Link)

- Best hybrid option in PaaS
- Near 100% compatibility with SQL Server
- Built-in high availability
- Integrates with on-premises via VPN / ExpressRoute

NO → Azure Arc enabled SQL (for multi-cloud / edge)

- Run SQL Server on-premises, other clouds, or edge
- Centralized management in Azure
- Consistent security and compliance

Why Choose Azure SQL (PaaS)?

- **Fully managed** – No infrastructure to manage
- **High availability** – Up to 99.99% SLA
- **Built-in security** – TDE, Threat detection, Advanced Data Security
- **Elastic scaling** – Scale compute & storage independently (vCore)
- **Automated backups &** point-in-time restore
- **Global scale** – Geo-replication & failover groups

Monitor & Optimize (All Models)

- Azure Monitor – Metrics, Alerts, Logs
- Query Performance Insight
- Automatic tuning (Index, Plan, Force Last Good Plan)
- Cost Management + Advisor Recommendations
- Scale up / down based on 70%+ resource utilization or storage > 80%

Quick Model Selection Guide

Small DBs, Dev/Test, ≤ 4 TB, predictable cost	DTU Model
Production, need transparency, SQL SA available	vCore – Provisioned + AHB
Intermittent, unpredictable workloads	vCore – Serverless
Very Large DBs, 100 TB scale, high IOPS	vCore – Hyperscale
Need SQL Agent, multiple DBs, minimal code changes	Managed Instance
Need full OS control, custom configs, 3rd-party software	SQL on Azure VM

KEY TAKEAWAY

Choose the right service and model based on your workload requirements, control needs, licensing strategy, and cost optimization goals.

- PaaS for Simplicity
- IaaS for Maximum Control
- Use AHB for Savings
- Monitor Continuously and Scale Proactively

Sources: Microsoft Learn, Azure SQL Documentation, Azure Pricing (May 2025) | Note: Pricing and limits are subject to change. Always validate in Azure Pricing Calculator.